

AUDIT SURFACE FOR

Capability Matters: A Casebook

Validation & Audit

INDEXES · PER-CASE REFERENCES

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Learning Engineering for Next-Generation Systems · v2.1

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SECTION

What this document is for

Validation and Audit is the auditor-facing companion to **Capability Matters: A Casebook**. It carries the surfaces an instructor, reviewer, or accreditation reader needs to verify the casebook — the indexes that let any case be located by domain or by LENS course, and the per-case reference appendix that lets any source be traced. It is the same data the casebook itself ships, lifted into a standalone document so the LENS Companion can stay focused on what LENS **is** (the five competencies, the LEOs, the course mapping) without the audit pages travelling with it.

The document is in three parts:

- I. Cases by primary domain — every case classified by the operational domain it primarily evidences, so a domain expert can find the cases that touch their territory without scanning the table of contents.
- II. Cases by LENS course — every case classified by the LENS course it serves as a worked example for, so syllabus authors and capstone advisors can find the cases that fit a course's stated learning outcomes.
- III. References, case by case — every reference cited in every case, organised by case in number order, with an explicit **Retrieved from:** line per entry so a reviewer can locate the canonical source.

On verification. The running per-case verification track lives in the repository at `casebook/verification-log.md`, a 194-row table whose six auto-prefilled columns track the structural and sourcing checks the build pipeline can answer on its own. The seventh column — **conclusions reasonable** — is the human case-by-case content read this document supports. The references appendix below is the per-case checklist for that read; the **Retrieved from:** lines are the canonical-access record each case acquires as the verification pass closes.

DOMAIN INDEX

The cases by domain.

The matrix at the front lists every case in number order. This index reorganises the cases by primary domain — a reader following healthcare, defense, education, or any other thread can find the relevant cases together. Each case appears once, under its primary domain. Where a section opens with a callout, it names the dataset’s standout failure and, where one exists in that domain, its standout success.

DEFENSE

STANDOUT FAILURE	Case 5	Operation Eagle Claw (1980). Cross-organizational capability that existed on no service’s deliverable list; the mission was improvised across the Army, Navy, Marines, Air Force, and CIA. Produced the Holloway Commission, USSOCOM, and Goldwater–Nichols.	
	Case 18	U.S. Nuclear Navy / Rickover Training Model (1954–). The longest continuous capability–engineering program in any high–consequence domain. Sixty–plus years of zero reactor accidents through training treated as a system parameter.	
124	USS Fitzgerald & USS John S. McCain	2017	T·K·N
126	F–35 Sustainment & Maintainer Shortage	ongoing	T·K·D
127	Military Fratricide — Desert Storm to Afghanistan	1991 – 2014	T·H·K
128	Operation Eagle Claw	1980	T·K
129	Patriot Missile / Dhahran	1991	D·H·K
130	USS Vincennes & Iran Air Flight 655	1988	H·T
131	Stanislav Petrov / 1983 False Alert	1983	H·T
133	Mark 14 Torpedo Failures	1941 – 1943	T·K·G
134	V–22 Osprey	1991 – present	T·H·N
135	9/11 Intelligence Sharing Failures	1996 – 2001	G·K
137	U.S. Nuclear Navy / Rickover Training Model	1954 – present	T·K·N
138	MIL–STD–1472H — Human Engineering as a Binding Acquisition Standard	2020 revision (series since 1968)	G·K

139	GIFT and the Adoption Gap	2012 – present	K·G·N
140	Navy Surface Warfare Readiness Reform	2018 – present	T·K·N
141	DARPA Digital Tutor — Compressing Years of IT Expertise into 16 Weeks	2009 – 2014	H·K·D

AVIATION

STANDOUT FAILURE

Case 7

Boeing 737 MAX / MCAS (2018–19). A single-string angle-of- attack sensor, a flight-control law that assumed pilot mental models from prior models, and a certification chain that did not knit the pieces together. 346 lives.

STANDOUT SUCCESS

Case 70

Crew Resource Management & CAST (1981–). First-officer authority engineered into the cockpit, a continuous learning architecture across the industry, and the safety record that followed.

96	AeroPerú Flight 603	1996	H·T
97	Boeing 737 MAX / MCAS	2018 – 2019	D·T·H
102	Air France Flight 447	2009	T·H
103	Kegworth / British Midland 92	1989	T·H·K
104	Asiana Airlines Flight 214	2013	T·H
105	Helios Airways Flight 522	2005	T·H
106	TransAsia Airways Flight 235	2015	T·H
107	Eastern Air Lines Flight 401	1972	H·T
109	Colgan Air Flight 3407	2009	T
110	Atlas Air Flight 3591	2019	T·K
112	NASA Saturn V Documentation	1972 – present	K
117	Crew Resource Management & CAST	1981 – present	T·H·N
118	Korean Air Safety Transformation	2000 – present	T·N
119	Aviation Safety Reporting System (ASRS)	1976 – present	T·K·N
120	EGPWS / TAWS — Closing the CFIT Category in Commercial Aviation	1996 – 2002	H·K·G
121	TCAS — Coordinated Collision Avoidance and the Überlingen Lesson (Version 7.1)	1981 – 2008 (TCAS II)	H·K·G
122	Singapore Airlines Safety Transformation	1980s – present	T·N
132	F-22 OBOGS Hypoxia — The Sustainment-Era Instrumentation Gap	2008 – 2012	H·K·N

HEALTHCARE

STANDOUT
FAILURE

Case 61

EHR / CPOE Implementation (2005–). Roughly \$30B federal investment in systems designed for billing and administration, deployed against clinical workflow. Usability is now itself a patient-safety variable at scale.

STANDOUT
SUCCESS

Case 109

WHO Surgical Safety Checklist (2008–). One page, nineteen items, paired with a required pause. Halved surgical mortality in the multi-site pilot; the smallest effective capability intervention in the dataset.

1	Therac-25	1985 – 1987	H·D·G
2	EHR / CPOE Implementation	2005 – present	H·D·G
4	Mid Staffordshire NHS Foundation Trust	2005 – 2009	G·N·K
5	Epic Sepsis Model	2017 – 2021	D·G·N
8	Medical Errors as Systemic Failure	1999 – present	T·H·N·K·G
9	Vioxx Withdrawal	1999 – 2004	G·D
11	California Nurse Staffing Ratios — Legislating a Capability Requirement	2004 – 2010	G·K
12	ACGME 80-Hour Resident Duty-Hour Reform	2003–2017	T·K·N
13	Implementation Science in Healthcare — The 17-Year Gap	ongoing	K·G·N
19	Keystone ICU / Pronovost Checklist	2004 – present	T·N
20	TREWS / Sepsis Watch	2018 – 2022	H·K·G
21	Sterile-Cockpit Ward Rounds — Adapting an Aviation Principle to Clinical Handoff	2024 – 2025	H·K·N
22	ChatGPT in Healthcare — Hallucination Cases	2023 – present	H·D
23	WHO Surgical Safety Checklist	2008 – present	T·N
24	Bristol Heart Babies Reform	1984 – present	G·N
32	Annual-Screening UI Redesign + CDS at University of Missouri Health Care	2020	T·D·N
33	Alert-Fatigue Redesign — Cutting EHR Alerts Without Cutting the Safety Signal	2019 – 2024	T·D·N
34	COMPOSER Sepsis Prediction — The Third Clinical-AI Sepsis Case	2022 – 2024	T·K·D
35	Radiology AI Miscalibration	2018 – present	H·K·D
36	AlphaFold — Protein Structure Prediction	2020 – present	T
37	DeepMind Mammography — High-Profile Nature Paper, Replicability Pushback	2020	T·K·D

39 TeamSTEPPS 2006 – present

T-N

EDUCATION AT SCALE

STANDOUT
FAILURE

Case 149

Summit Learning / Personalized Learning Rollout (2014–19). A defensible pedagogical platform, well funded, deployed across 380 schools — and pulled by parent revolt because the governance architecture (consent, evidence, exit) was never engineered alongside it.

STANDOUT
SUCCESS

Case 110

Georgia State University Predictive Analytics (2012–). 800 risk factors per student, advising triggered by early warning signs, equity as a primary constraint. Six-year graduation 32% → 54%; equity gaps eliminated.

45 Tennessee Voluntary Pre-K Study 2009–2018

G-D

46 Algorithmic Bias in Educational Predictive Analytics ongoing

G-H-D

51 Atlanta Public Schools Cheating Scandal 2009 – 2015

G-N

53 inBloom 2014

G

54 Summit Learning / Personalized Learning Rollout 2014–2019

G-T-K

67 Cognitive Tutor / Carnegie Learning 1990s – present

T

80 Georgia State University Predictive Analytics 2012 – present

T-K

142 xAPI / Total Learning Architecture — Interoperability Gap ongoing

K-G

205 The Discipline We Build Next ongoing

T-K-N-G-H-D

ENERGY

STANDOUT
FAILURE

Case 69

Three Mile Island (1979). Training matched the design-basis events the engineers had imagined; the accident was not one of them. The control-room interface confused command signal with valve state. Catalysed industry-wide reform.

STANDOUT
SUCCESS

Case 172

INPO and the Nuclear Academy (1979–). The industry built the institution TMI revealed it needed. No INES-level event at U.S. commercial reactors since.

146 Northeast Blackout 2003

H-K

GOVERNMENT

STANDOUT
FAILURE

Case 162

Australia Robodebt (2016–20). An income-averaging algorithm that reversed the burden of proof onto welfare recipients. The Royal Commission found “venality, incompetence and cowardice”; the scheme was linked to deaths, including suicide.

NOTE

Case 111

No paired-intervention success in the government dataset. The reforms that follow these cases — Royal Commissions, statutory restructures — are documented in the narratives but did not produce a LENS-style success case in time for this volume.

7 VA Wait-Time Scandal 2014

G-K-N

INDUSTRIAL

STANDOUT
FAILURE

Case 147

Bhopal (1984). Training, knowledge, normalisation of deviance, and governance all collapsed in the same plant. Roughly 4,000 immediate deaths; chronic harms continue. The cautionary case for outsourced operational responsibility.

STANDOUT
SUCCESS

Case 73

Toyota Production System / Andon Cord (1950s–). Any operator can stop the line. The smallest authority-architecture intervention in the dataset, and the most durably copied.

144 Kodak Digital Camera Stagnation 1975–2012

D-K-N

147 Takata Airbag Inflators 2008 – 2023

D-G

149 Volkswagen Dieselgate 2015

D-G

151 GM Ignition Switch 2002 – 2014

D-G

155 Toyota Production System / Andon Cord 1950s – present

N-G

TECHNOLOGY

STANDOUT
FAILURE

Case 66

UK Post Office Horizon Scandal (1999–2015). Institutional refusal to believe sub-postmasters’ reports of computer error, sustained over fifteen years. Hundreds of wrongful convictions; ruined lives and at least four suicides — the longest-running operator-feedback failure in the dataset.

NOTE

Case 132

The technology dataset’s success story is filed under Healthcare: AlphaFold (Case 36), the contemporary worked example of computational capability-engineering done as a discipline.

143 Knight Capital Trading Loss 2012

D-K

148	Wells Fargo Fake Accounts	2011 – 2016	G·N
152	TSB Bank IT Migration	2018	H·G
153	Equifax Data Breach	2017	G·K
157	AI-Augmented Coding Tools	2021 – present	T·H

AI IN EDUCATION

88	LiveHint AI — Evaluating an AI Tutor for Bias Across Foundation Models	2025	T·K·N
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K-12 EDUCATION

48	School Surveillance and Black Student Outcomes — Infrastructure as the Mechanism	2010s – 2022	G·K·N
60	Houston EVAAS — Value-Added Teacher Evaluation on Trial	2011–2017	G
61	Sold a Story — The Science of Reading and the Decades-Long Research-Practice Gap	1997–2023	K·T
62	Gates Intensive Partnerships — Measurement Without a Coupled Lever	2009–2018	G·K
63	LAUSD iPad Program — The Common Core Technology Project	2013–2015	D·H·G
66	Growth-Mindset National Experiment — Heterogeneous Effects	2019	D·N·K
75	Chen et al. — Rural China AI Devices and the Equity-Direction Finding	2025	T·K·D
95	PBIS Implementation Fidelity — Why the Framework Travels Only with Its Measurement Loop	1997–2024	T·G

K-12 MATHEMATICS

72	ASSISTments — National Replication and Long-Term Follow-Through	2014 – present	T·K·D
84	Cognitive Tutor Algebra I at Scale — Year-One Null, Year-Two Positive	2007 – 2014	T·K·D

K-12 SCIENCE

74	Zhang/Scardamalia — Knowledge Building Across Three Cohorts	2009	T·K·D
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AIR TRAFFIC MANAGEMENT

116	Eurocat ATM Pilot Modernization — Small-Tier Verification as the Gateway to Big-Tier Change	2005	G·K·H
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ANESTHESIOLOGY

27	Anesthesia Monitoring Standards and the APSF — The Mortality Decline	1986 – present	H·K·G
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AVIATION INFRASTRUCTURE

115	FAA NextGen and the ADS-B Out Transition	2003 – 2020	G·D·K
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AVIATION SAFETY

114 FAA Aging-Aircraft Program — Widespread Fatigue Damage and the Part 26 Rule 1988 – 2010s **G·D·K**

123 FSF CFIT and ALAR Task Forces — Industry-Level Institution Building After a Spike 1992 – 2000s **G·K·N**

CLINICAL CARE

15 I-PASS Handoff Bundle — Structuring the Human-to-Human Transfer 2014 **H·K·N**

29 Bar-Code Medication Administration — Cue at the Point of Care 2010 **H·K·D**

CLINICAL MEDICINE

6 Racial Bias in Pain Assessment — The False-Belief Mechanism 2016 **T·K·N**

25 Removing the Race Coefficient from eGFR 2021 **D·G·N**

CLINICAL/TRANSLATIONAL RESEARCH

40 Team Science Training for Clinical and Translational Scientists 2020 – 2022 **K·N**

CORPORATE L&D

65 Training Transfer — The Gap Between Learning and Doing 2010 **K·N**

70 High-Impact Learning System — Engineering the Environment, Not Just the Event 2001 – present **K·N**

79 The Kirkpatrick Chain of Evidence — Where Corporate L&D Stops 1959 – present **K·N**

83 Brinkerhoff Success Case Method — Tails as the Evaluation Instrument 2005 – present **K·N**

ED-TECH

47 Algorithmic Bias in Automated Exam Proctoring 2022 **D·N·K**

EDUCATION (NORWAY)

92 Norway's National Expert Commission on Learning Analytics 2021 – 2023 **G·K·N**

EDUCATION AT SCALE

50 Wisconsin DEWS — A Decade of Algorithmic Dropout Prediction 2012 – 2024 **D·K·N**

69 Duolingo Half-Life Regression — Spaced Repetition at Consumer Scale 2016 **T·D**

GENERAL AVIATION

100 Glass-Cockpit Transition in Light Aircraft — Technology Outran Training 2010 **H·N·K**

GLOBAL HEALTH

18	PEPFAR HIV Training Across 16 Sub-Saharan African Countries — Modality Comparison Under Disruption 2023	K·N·D
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43	Rwanda mHealth Maternal Care — Community Health Workers as the Capability Interface 2013 – 2018	H·N
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GOVERNMENT

49	UK Ofqual A-Level Algorithm — National-Scale Grading Replaced by Algorithm, Withdrawn in Days 2020	D·K·N
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HEALTH PROFESSIONS EDUCATION

28	Interprofessional Education and the Evidence Gap 2013 – 2015	K·N
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HEALTHCARE/ONCOLOGY

3	Watson for Oncology — Delegation Marketed Ahead of Capability 2013 – 2018	D·K·N
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HIGHER EDUCATION

55	Enrollment-Algorithm Yield Optimization Across U.S. Higher Education (Brookings synthesis 2021) 2010s – present	T·K·N
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57	GAO Online Program Manager Oversight Gap (GAO-22-104463) 2022	D·K·N
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58	USC × 2U Online MSW — When the Delegation Becomes the Product (Luna v. USC) 2010s – 2024	D·K·N
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85	OU Analyse — Predictive Learning Analytics and Teacher Use at Scale 2019 – 2023	T·K·D
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86	Gándara — Detecting and Mitigating Algorithmic Bias in College Student-Success Prediction 2024	D·K·N
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90	Algorithmic College Admissions — Vendors' Claims vs. Applicants' Perceptions 2025	T·K·N
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HIGHER EDUCATION (AFRICA)

89	Data Privacy and Learning Analytics on the African Continent 2022	K·G·N
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HIGHER EDUCATION (BRAZIL)

82	MMALA — A Maturity Model for Responsible Learning-Analytics Adoption (Brazil) 2024	K·N
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HIGHER EDUCATION (LATIN AMERICA)

91	LALA — Building Learning-Analytics Governance Capacity Across Latin America 2017 – 2020	K·N
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HIGHER ED (UK)

81	Open University 'Ethical Use of Student Data' and OU Analyse 2014 – 2025	G·K·N
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HIGHER-ED ANALYTICS

52 Purdue Course Signals — The Reverse-Causality Retention Claim 2012 – 2013 **D-K-N**

HOSPITAL PHARMACY

42 Australian Hospital-Pharmacy Technician Role Redesign 2016 **D-N-H**

IMPLEMENTATION SCIENCE

41 Implementation-Science Training — Stated Goals Outrunning Operational Practice 2020s **K-N**

LEARNING ANALYTICS

73 The Doer Effect at Scale — Replication, AI-Generated Questions, Non-WEIRD Extension 2016 – 2025 **T-K-D**

87 Yu / Lee / Kizilcec — Protected Attributes in Learning-Analytics Models 2021 – 2024 **D-K-N**

94 Learning Analytics on the African Continent — A Scoping Review as the Present-State Map 2022 **K-N**

MEDICAL DEVICES

26 Pulse Oximetry Across Skin Tones 1990 – 2020 – 2025 **D-G-N**

MEDICAL EDUCATION

14 Barsuk SBML — Simulation-Based Mastery Learning Dissemination from Northwestern to the VA 2009 – 2020s **T-K**

17 Spaced Education RCTs in Medical Training 2007 – 2009 **H-K-D**

MEDICAL-DEVICE REGULATION

44 Japan PMDA — Pre-Approved Change Management as Regulatory Architecture for AI/SaMD 2014 – present **G-N**

MULTIMODAL LEARNING ANALYTICS

76 Multimodal Learning Analytics In-the-Wild — A First-Person Lessons-Learned Account 2023 **T-K-N**

NEUROSCIENCE/CONNECTOMICS

78 CIRCUIT / MICrONS — The Human Correction Layer at Petabyte Scale 2017 – present **H-K-N**

NUCLEAR ENGINEERING

156 INL / LWRs Control-Room Modernization — Sustainment Research for an Aging Fleet 2010 – present **D-H-K**

NUCLEAR POWER

154	INL Turbine-Control Upgrade — Low-Burden Cutover as a Human-Factors Finding	2014	G-K-H
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NURSING EDUCATION

16	NCSBN National Simulation Study — Licensing the 50% Substitution Rule	2014	G-K-D
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PHARMA SAFETY

38	iPLEDGE Isotretinoin REMS — When the Authorization Mechanism Underperforms	2006 – 2011	G-D-N
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SOFTWARE ENGINEERING EDUCATION

71	Reflective Practice on a Work-Based Software Engineering Program — Longitudinal Capability Development	2025 (preprint)	K-N
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SPACE / AEROSPACE

STANDOUT FAILURE	Case 140	Challenger & Columbia (1986 / 2003). Normalisation of deviance across two decades and two crews, captured in the Rogers and CAIB reports as a culture that accepted flying with flaws when problems were defined as normal and routine.
NOTE	Case 18	No paired-intervention success in the aerospace dataset. See Defense (Rickover, Case 137) for the safety-engineering counterpart that aerospace did not build.

98	Mars Climate Orbiter — Unit Mismatch	1999	D-K
101	Ariane 5 Flight 501	1996	D-K-H
108	NASA EVA-23 — Water Intrusion Inside a Spacesuit	2013	H-N-D
111	Challenger & Columbia	1986 / 2003	N-K-G
113	Boeing Starliner	2019 – 2024	K-D

SURGERY

30	Surgical Skill Video Peer-Rating Predicts Complications	2013	H-D-K
31	Language of Surgery / JIGSAWS — Decomposing Skill into Measurable Units	2009 – 2016	T-K-H

TUTORING

77	Hybrid Human-AI Tutoring — Augmentation, Not Delegation	2024	T-K-D
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WORKFORCE DEV

68	CIRCUIT — A Scalable, Equity-Centered Research Workforce Model	2017 – 2023 (six cycles)	T-K
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93 Singapore SkillsFuture — National Workforce Capability at Scale 2015 – present

G·K·D

About this index. Each case is filed under its primary domain, so every case appears exactly once. Cases that span several domains (a healthcare-AI case is also a technology case, a defense-training case is also an education case) are cross-referenced in their narratives. The standout failures and successes are editorial picks — the case in that domain the editors return to most often in studio.

The cases by LENS course.

Each case names the LEN courses it serves as a worked example for. This index inverts that mapping: for every course in the specialization, the cases that inform it, in case-number order. A case appears under every course it supports, so many appear more than once.

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37 cases

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References, case by case.

Every reference cited in every case, organised by case in number order. The intent is that any reader can locate every source the book draws on. Each entry preserves the in-case citation form (author, year, venue) and adds an explicit **Retrieved from:** line pointing to the canonical access path — a publisher URL, a DOI, an agency landing page, or a stable archival locator. Where that line is left blank, the source is under verification; the running status of that work is in casebook/verification-log.md. Future editions will fill in remaining locators as the per-case review closes them out.

Style. References are rendered in the form they appeared in the case — APA-style author/year/title/venue, with italicised book titles and journal names — supplemented by the explicit retrieval locator. Government and agency reports are cited by report number when one exists. Trade-book references that carry no DOI use ISBN as the stable identifier; that mapping is on the verification log.

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2024 – 2025

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CASE 30 Surgical Skill Video Peer-Rating Predicts Complications

2013

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CASE 31 Language of Surgery / JIGSAWS — Decomposing Skill into Measurable Units

2009 – 2016

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CASE 32 Annual-Screening UI Redesign + CDS at University of Missouri Health Care

2020

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CASE 33

Alert-Fatigue Redesign — Cutting EHR Alerts Without Cutting the Safety Signal

2019 – 2024

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CASE 34

COMPOSER Sepsis Prediction — The Third Clinical-AI Sepsis Case

2022 – 2024

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CASE 35

Radiology AI Miscalibration

2018 – present

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2020 – present

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CASE 37 DeepMind Mammography — High-Profile Nature Paper, Replicability Push-back

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